Putnam - Solutions

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Chapter 1

Putnam 1938

A1

Let $A(k) = ak^3 + bk^2 + ck + d$, where $d = M$ since $A(0) = M$. Let us compute the volume by integrating $A(k)$:

$$V = \int_{-\frac{h}{2}}^{\frac{h}{2}} A(k)dk$$

$$= \left[\frac{a}{4}k^4 + \frac{b}{3}k^3 + \frac{c}{2}k^2 + Mk\right]_{-\frac{h}{2}}^{\frac{h}{2}}$$

$$= \frac{bh^3}{12} + Mh$$

Our goal is to get

$$V = \frac{h(B + 4M + T)}{6}$$

Notice that

$$B = A\left(-\frac{h}{2}\right) = -a\frac{h^3}{8} + b\frac{h^2}{4} - c\frac{h}{2} + M$$

$$T = A\left(\frac{h}{2}\right) = a\frac{h^3}{8} + b\frac{h^2}{4} + c\frac{h}{2} + M$$

thus

$$B + T = \frac{bh^2}{2} + 2M$$

Substituting into our goal, we simplify

$$V = \frac{h(B + 4M + T)}{6} = \frac{h(\frac{bh^2}{2} + 6M)}{6} = \frac{bh^3}{12} + Mh$$

Hence, the volume formula is indeed correct.

We can use this formula to derive the formula for a cone and sphere. For a cone with base radius r and height h, the area formula is

$$A(k) = \pi \left(\frac{r}{h} \left(k + \frac{h}{2} \right) \right)^2$$

thus a 2nd degree polynomial, which satisfies the requirements for using the special formula, hence

$$V = \frac{h(B + 4M + T)}{6}$$

$$= \frac{h(\pi r^2 + 4\pi \left(\frac{r}{2}\right)^2 + 0)}{6}$$

$$= \frac{h(2\pi r^2)}{6} = \frac{\pi r^2 h}{3}$$

For a sphere with radius $r = \frac{h}{2}$, the cross-sectional area is

$$A(k) = \pi \left(\frac{h^2}{4} - k^2 \right)$$

thus again a 2nd degree polynomial in k, so we apply the special formula to get

$$V = \frac{h(B + 4M + T)}{6}$$

$$= \frac{h(0 + 4\pi \frac{h^2}{4} + 0)}{6} = \frac{\pi h^3}{6}$$

Expressed in radius $2r = h$, we get

$$V = \frac{4}{3}\pi r^3$$

A2

$$V = 2 \cdot \frac{\pi r^2 h}{3} + \pi r^2 h = \frac{5}{3}\pi r^2 h$$

$$A = 2\pi r h + 2\pi r \sqrt{r^2 + h^2}$$

We want to express either r or h in terms of A and the other:

$$A - 2\pi r h = 2\pi r \sqrt{r^2 + h^2}$$

$$A^2 - 4A\pi r h + 4\pi^2 r^2 h^2 = 4\pi^2 r^2 (r^2 + h^2)$$

$$A^2 - 4A\pi r h + 4\pi^2 r^2 h^2 = 4\pi^2 r^4 + 4\pi^2 r^2 h^2$$

$$A^2 - 4A\pi r h = 4\pi^2 r^4$$

$$h = \frac{A^2 - 4\pi^2 r^4}{4A\pi r}$$

Substituting into the volume equation, we get

$$V = \frac{5}{3}\pi r^2 \frac{A^2 - 4\pi^2 r^4}{4A\pi r}$$

$$= \frac{5}{12}\frac{r(A^2 - 4\pi^2 r^4)}{A}$$

We want to maximize this equation given a fixed A, so we can ignore the coefficient and denominator, leaving only

$$r(A^2 - 4\pi^2 r^4) = A^2 r - 4\pi^2 r^5$$

Setting the derivative to zero, we get

$$A^2 - 20\pi^2 r^4 = 0$$

$$\Rightarrow r = \sqrt{\frac{A}{\sqrt{20}\pi}}$$

Note that, from the height equation, we know that

$$0 < A^2 - 4\pi^2 r^4$$

$$\Rightarrow r < \sqrt{\frac{A}{2\pi}}$$

$\sqrt{20} > \sqrt{16} = 4 > 2$, so the r we found is safely within the bounds.

A3

Velocities are

$$\frac{dx}{dt} = 3t^2 - 1$$

$$\frac{dy}{dt} = 4t^3 + 1$$

I think they are asking for the speed, which is

$$\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = \sqrt{16t^6 + 9t^4 + 8t^3 - 6t^2 + 2}$$

Since we are finding the maximum, we can ignore the square root. Since a polynomial is differentiable everywhere, its maxima happen when derivative is zero

$$96t^5 + 36t^3 + 24t^2 - 12t = 0$$

Clearly, $t = 0$ is an extrema (or saddle). To know whether it's a maximum, we can take the second derivative at 0

$$480t^4 + 108t^2 + 48t - 12$$

which equals -12 (negative) when t are replaced with 0, hence $t = 0$ is a local maximum.

To know whether it's an inflection point, we can compute $\frac{d^2y}{dx^2}$ at $t = 0$, which should be possible since the tangent is not vertical (specifically, $\vec{v}(0) = (-1, 1)$):

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{3t^2 - 1}{4t^3 + 1}$$

$$\frac{d}{dx}\frac{dy}{dx} = \frac{d}{dt}\frac{dy}{dx} \cdot \frac{dt}{dx} = \frac{d}{dt}\frac{dy}{dx} \cdot \frac{1}{\frac{dx}{dt}}$$

$$= \frac{(4t^3 + 1)(6t) - (3t^2 - 1)(12t^2)}{(4t^3 + 1)^2} \cdot \frac{1}{3t^2 - 1}$$

$$= \frac{-12t^4 + 12t^2 + 6t}{(4t^3 + 1)^2(3t^2 - 1)}$$

Substituting 0, the whole thing becomes 0. Hence, the second derivative is zero, but we still need to know whether there is a sign change for the second derivative (which represents change in concavity).

For t near 0, the denominator is always negative. However, the derivative of the numerator is 6 (positive) at $t = 0$, and since the value of the numerator is 0 at $t = 0$, the numerator changes sign when $t < 0$ becomes $t > 0$. Hence, there is a sign change in the second derivative at $t = 0$, hence it's an inflection point.

A4

I don't understand the problem.

A5

(1) Using L'Hopital's,

$$\lim_{x\to\infty} \frac{x^2}{e^x} = \lim_{x\to\infty} \frac{2x}{e^x} = \lim_{x\to\infty} \frac{2}{e^x} = 0$$

Alternatively, we can argue exponential growth dominates polynomial growth.
(b) Since

$$\lim_{x\to 0} (1 + \sin 2x)^{\frac{1}{x}}$$

$$= e^{\lim_{x\to 0} \frac{1}{x} \log(1+\sin 2x)}$$

with L'Hopital's,

$$= e^{\lim_{x\to 0} \frac{2\cos 2x}{1+\sin 2x}} = e^2$$

is finite. Inside $[0, \infty)$, it's the only undefined point of the integrand, which has a finite limit, hence the integral is bounded for every k. Hence, as $k \to 0$, the integral goes to 0. Therefore, we can apply L'Hopital's

$$\lim_{k\to 0} \frac{\int_0^k (1+\sin 2x)^{\frac{1}{x}} dx}{k}$$

$$= \lim_{k\to 0} (1+\sin 2k)^{\frac{1}{k}}$$

which we found to be equal to e^2.

A6

Suppose the pool is a 1x1 square. Let s, r be swimming and running speeds respectively. If $s > r$, then the obvious solution is to swim diagonally directly to the opposite corner. If $s < r$, there are two cases.

If swimming the hypothenuse is slower than running both edges, i.e. $\frac{\sqrt{2}}{s} > \frac{2}{r} \Rightarrow s < \frac{r}{\sqrt{2}}$, then the solution is to run along the edges.

If she starts swimming at the start, there are three options. Reaching the bottom or left edges, reaching the top or right edges, or reaching the opposite corner. The latter, by our initial assumption $\frac{\sqrt{2}}{s} > \frac{2}{r}$, is obviously slower. Obviously, there's a fourth option to reach somewhere else in the water, but she will eventually reach one of those three options, and this fourth option checkpoint would simply be a point in her path to one of those three options.

If she lands on the bottom or left edges, then she effectively used a longer path than simply running directly to the landing point. Hence, this option is discarded.

If she lands on the top or right edges, she has many options. Going back to the starting point or to the bottom or left edge is stupidly wrong. The remaining options are landing back onto the top or right edge, or simply reaching the opposite corner. Re-landing onto the top or right edge is, again, a longer path than simply running along the edge. Hence, her only viable option is running directly to the opposite corner.

Depending on her landing ordinate $0 < y < 1$ on the right edge (the reasoning is equivalent for the top edge), her time is

$$T = \frac{\sqrt{1+x^2}}{s} + \frac{1-x}{r} > \frac{\sqrt{1+x^2}}{\frac{r}{\sqrt{2}}} + \frac{1-x}{r}$$

$$= \frac{1-x+\sqrt{2}\sqrt{1+x^2}}{r}$$

The numerator is always greater or equal to 2, because

$$\begin{aligned}\sqrt{2}\sqrt{1+x^2} - x + 1 &\geq 2 \\ \Rightarrow \sqrt{2}\sqrt{1+x^2} &\geq x + 1 \\ \Rightarrow 2(1+x^2) &\geq x^2 + 2x + 1 \\ \Rightarrow x^2 + 1 &\geq 2x \\ \Rightarrow (x-1)^2 &\geq 0\end{aligned}$$

is true.

Hence, $T > \frac{2}{r}$, thus running along the edges is always faster if swimming the hypothenuse is slower than running the edges, if she starts by swimming.

Another case is when $\frac{r}{\sqrt{2}} \leq s < r$, i.e. swimming the hypothenuse is faster than running the edges, but running speed is still higher than swimming speed. Going back to our three options,

A7

(1) Simple double integration exercise.
(2) Using the symmetry of the surface $z = xy$, we can reduce to three cases for the line, when viewed from above:

- passes through the center
- crosses either x or y axis
- crosses both x and y axes

Starting with the first case, if the line passes through the center, then $y = kx$ for some k, whence $z = xy = x(kx) = kx^2$. This is only a line when $k = 0$, where we end up with the line $y = z = 0$ varying with x. The only case missing is the vertical line case, where $x = 0$. This is obviously a line, since $z = xy = 0$. Hence $x = z = 0$ with only y varying, which is a line.

For the second case, if a line crosses only one axis, then it is parallel with that axis (when viewed from above). Suppose $x = x_0$ is fixed, then $z = x_0 y$, which is a line with slope x_0 and independent variable y. Same thing when y is fixed. Hence, we have an infinite set of such lines. The first case is actually a special case, where $x_0 = 0$ or $y_0 = 0$.

Finally, if a line crosses both the x and y axes, then $x = 0$ and $y = 0$ reach occur once, whence $z = xy = 0$ at those two points, hence the line must lay flat against the horizontal plane, i.e. $z = 0$ everywhere. However, everywhere else the line passes through, the plane assumes value $z \neq 0$, hence it cannot be a line.

To summarize, we have the lines:

- $z = x_0 y$, $x = x_0$ for $x_0 \in \mathbb{R}$
- $z = y_0 x$, $y = y_0$ for $y_0 \in \mathbb{R}$

B1

(1) Solution found online. Consider the cofactor matrix A_{ij} and the transpose of the original matrix A^T.

$$\begin{bmatrix} A_{11} & A_{12} & A_{13} & A_{14} \\ A_{21} & A_{22} & A_{23} & A_{24} \\ A_{31} & A_{32} & A_{33} & A_{34} \\ A_{41} & A_{42} & A_{43} & A_{44} \end{bmatrix} \begin{bmatrix} a_{11} & a_{21} & a_{31} & a_{41} \\ a_{12} & a_{22} & a_{32} & a_{42} \\ a_{13} & a_{23} & a_{33} & a_{43} \\ a_{14} & a_{24} & a_{34} & a_{44} \end{bmatrix}$$

If we compute the matrix product, we notice that

$$a_{i1}A_{1i} + a_{i2}A_{2i} + a_{i3}A_{3i} + a_{i4}A_{4i} = \det A = d$$

but

$$a_{i1}A_{1j} + a_{i2}A_{2j} + a_{i3}A_{3j} + a_{i4}A_{4j} = 0$$

for $i \neq j$, because it's equivalent to the determinant of A where the j-th row is replaced with the i-th row. Hence, the resulting matrix is

$$\begin{bmatrix} d & 0 & 0 & 0 \\ 0 & d & 0 & 0 \\ 0 & 0 & d & 0 \\ 0 & 0 & 0 & d \end{bmatrix}$$

which has determinant d^4. Since it was the result of multiplying the transpose with the cofactor matrix, and the transpose has determinant d, then the cofactor matrix itself has determinant $\frac{d^4}{d} = d^3$.
(2) If $P(x_1) = x_1$, then $P(P(x_1)) = P(x_1) = x_1$, hence $P(P(x_1)) = x_1$, so x_1 is a root of $P(P(x)) = x$. Same for the other root x_2.

I'm too dumb for the rest.

B2

Solution found online.

$$\begin{aligned} yy'' &= 2y'y' \\ \Rightarrow \frac{y''}{y'} &= 2\frac{y'}{y} \\ \Rightarrow \log y' &= 2\log y + C \\ \Rightarrow y' &= Cy^2 \\ \Rightarrow C\frac{dy}{y^2} &= dx \\ \Rightarrow -\frac{C}{y} &= x + D \end{aligned}$$

Since it passes through $x = 1, y = 1$, then $-C = 1 + D$. Hence, we have the family of solutions

$$\frac{1+D}{x+D} = y$$

B3

Using the disk's reference frame, the insect is simply turning at a constant speed and advancing at a constant speed. Hence, its path is an arc of a bigger circle. The disk's edge and the big circle intersect at right angles, hence the insect's path touches the disk's edge at right angle. Since at the end of its path, the insect is facing north, the insect touches the edge again at the disk's northernmost point.

B4

Using algebra, we can find the parabola's equation:

$$x^2 + \left(\frac{m}{2} - y\right)^2 = \left(\frac{m}{2} + y\right)^2$$

$$x^2 + y^2 - my + \frac{m^2}{4} = y^2 + my + \frac{m^2}{4}$$

$$y = \frac{x^2}{2m}$$

assuming the parabola passes through the origin. The derivative is thus

$$y' = \frac{x}{m}$$

This is the slope of the tangent line. For the perpendicular line, the slope is $-\frac{m}{x}$. By symmetry, we can restrict ourselves to using $x < 0$, hence the lines have positive slope $\frac{m}{x_0}$ where x_0 is the horizontal distance from the origin. Using algebra, we can find the line's equation:

$$b = \frac{m}{x_0} \cdot x_0 + \frac{x_0^2}{2m} = m + \frac{x_0^2}{2m}$$

$$\Rightarrow y = \frac{m}{x_0}x + m + \frac{x_0^2}{2m}$$

Equating with the parabola, we get

$$\frac{m}{x_0}x + m + \frac{x_0^2}{2m} = \frac{x^2}{2m}$$

$$0 = \frac{x^2}{2m} - \frac{m}{x_0}x - m - \frac{x_0^2}{2m}$$

$$= x_0x^2 - 2m^2x - 2m^2x_0 - x_0^3$$

The quadratic formula gives

$$x = \frac{m^2}{x_0} + \sqrt{\frac{m^4}{x_0^2} + 2m^2 + x_0^2}$$

which can be simplified

$$x = \frac{m^2}{x_0} + \sqrt{\left(\frac{m^2}{x_0} + x_0\right)^2} = \frac{2m^2}{x_0} + x_0$$

The corresponding y value is

$$\frac{2m^3}{x_0^2} + 2m + \frac{x_0^2}{2m}$$

The distance is

$$\sqrt{\left(\frac{2m^3}{x_0^2} + 2m + \frac{x_0^2}{2m} - \frac{x_0^2}{2m}\right)^2 + \left(x_0 + \frac{2m^2}{x_0} + x_0\right)^2}$$

$$= \sqrt{\left(\frac{2m^3}{x_0^2} + 2m\right)^2 + \left(2x_0 + \frac{2m^2}{x_0}\right)^2}$$

$$= \sqrt{\frac{4m^6}{x_0^4} + \frac{12m^4}{x_0^2} + 12m^2 + 4x_0^2}$$

$$= \frac{2}{x_0^2}\sqrt{m^6 + 3m^4x_0^2 + 3m^2x_0^4 + x_0^6}$$

$$= \frac{2}{x_0^2}(m^2 + x_0^2)^{\frac{3}{2}}$$

We can thus minimize it with respect to some $z = x_0^2$:

$$\frac{(m^2 + z)^{\frac{3}{2}}}{z}$$

The derivative is

$$\frac{\frac{3}{2}z(m^2 + z)^{\frac{1}{2}} - (m^2 + z)^{\frac{3}{2}}}{z^2}$$

Setting it to zero, we get

$$\frac{3}{2}z(m^2 + z)^{\frac{1}{2}} = (m^2 + z)^{\frac{3}{2}}$$

$$\frac{3}{2}z = m^2 + z$$

$$z = 2m^2$$

Plugging it back into the distance, we get a minimum distance of

$$\frac{2}{2m^2}(m^2 + 2m^2)^{\frac{3}{2}} = \frac{(3m^2)^{\frac{3}{2}}}{m^2} = \sqrt{27}m = 3\sqrt{3}m$$

B5

The hyperbola's equation has the form

$$x^2 - y^2 = c$$

assuming it's horizontal. The vertical hyperbola would simply exchange x and y, but the solution is similar.

The derivative is

$$\frac{dy}{dx} = \frac{x}{y}$$

The tangent's y-intercept can be found for a given x_0

$$b = y_0 - \frac{x_0^2}{y_0}$$

Let D be the distance from the line to the origin. We can solve for D using the fact that the right triangle formed by the tangent, y-axis, and distance, has the same legs proportion as the tangent's slope:

$$D^2 + \left(\frac{x_0}{y_0} D\right)^2 = b^2$$

$$D^2 = \frac{\left(y_0 - \frac{x_0^2}{y_0}\right)^2}{1 + \frac{x_0^2}{y_0^2}}$$

$$= \frac{\left(y_0^2 - x_0^2\right)^2}{y_0^2 + x_0^2}$$

Let h be the segment representing the distance from the tangent to the origin. For a given counterclockwise angle θ for h from the positive x-axis, the tangent's slope is

$$-\frac{\cos\theta}{\sin\theta}$$

which must correspond to the derivative, hence

$$\frac{x_0}{y_0} = -\frac{\cos\theta}{\sin\theta}$$

$$x_0^2 \sin^2\theta - y_0^2 \cos^2\theta = 0$$

We can derive the appropriate polar formulas for x_0^2 and y_0^2:

$$x_0^2 \sin^2\theta - (x_0^2 - c)\cos^2\theta = 0$$

$$x_0^2 = -\frac{c\cos^2\theta}{\sin^2\theta - \cos^2\theta}$$

and

$$(y_0^2 + c)\sin^2\theta - y_0^2\cos^2\theta = 0$$

$$y_0^2 = -\frac{c\sin^2\theta}{\sin^2\theta - \cos^2\theta}$$

We can rewrite the distance formula using these polar coordinates

$$D^2 = \frac{\left(-\frac{c\sin^2\theta}{\sin^2\theta-\cos^2\theta} + \frac{c\cos^2\theta}{\sin^2\theta-\cos^2\theta}\right)^2}{-\frac{c\sin^2\theta}{\sin^2\theta-\cos^2\theta} - \frac{c\cos^2\theta}{\sin^2\theta-\cos^2\theta}}$$

$$= \frac{\left(\frac{c\cos 2\theta}{-\cos 2\theta}\right)^2}{-\frac{c}{-\cos 2\theta}}$$

$$= \frac{c^2}{\frac{c}{\cos 2\theta}}$$

$$= c\cos 2\theta$$

Hence, the equation of the locus is

$$r = \sqrt{c\cos 2\theta}$$

which remains undefined for $\theta \in [\frac{\pi}{4}, \frac{3\pi}{4}] \cup [\frac{5\pi}{4}, \frac{7\pi}{4}]$, to match the fact that the tangent is impossible for those angles by graphical inspection. Basically, it's a figure 8 without the center.

Chapter 2

Putnam 1939

A1

Graph is symmetric above and below the x-axis, but the positive gradient is only found on the top. Hence, we have

$$y = \sqrt{x^3} = x^{\frac{3}{2}}$$

The derivative is

$$y' = \frac{3}{2}\sqrt{x}$$

which must be equal to 1, hence

$$\frac{3}{2}\sqrt{x} = 1$$

$$\Rightarrow x = \left(\frac{2}{3}\right)^2 = \frac{4}{9}$$

The path integral formula is

$$\int_0^{\frac{4}{9}} \sqrt{1 + (y')^2}dx$$

which we substitute and solve

$$\int_0^{\frac{4}{9}} \sqrt{1 + \left(\frac{3}{2}\sqrt{x}\right)^2}\,dx$$

$$= \int_0^{\frac{4}{9}} \sqrt{1 + \frac{9}{4}x}\,dx$$

$$= \left[\frac{4}{9} \cdot \frac{2}{3}\left(1 + \frac{9}{4}x\right)^{\frac{3}{2}}\right]_0^{\frac{4}{9}}$$

$$= \frac{8}{27}\left(1 + \frac{9}{4} \cdot \frac{4}{9}\right)^{\frac{3}{2}} - \frac{8}{27}$$
$$= \frac{8}{27}\left(2\sqrt{2} - 1\right)$$

A2

Derivative is

$$y' = 3x^2$$

The tangent's equation can be found

$$b = y_0 - x_0 \cdot y_0' = x_0^3 - x_0 \cdot 3x_0^2 = -2x_0^3$$
$$y = 3x_0^2 \cdot x - 2x_0^3$$

The intersections with the cubic can be found

$$3x_0^2 \cdot x - 2x_0^3 = x^3$$
$$0 = x^3 - (3x_0^2)x + 2x_0^3$$
$$= (x - x_0)(x - x_0)(x + 2x_0)$$

We already know it intersects at $x = x_0$, hence the desired point is $x = -2x_0$. The derivative at that point equals

$$y' = 3(-2x_0)^2 = 4(3x_0^2)$$

which is exactly four times the derivative $3x_0^2$ at $x = x_0$.

A3

We can express the cubic as a product using its roots

$$y = (x - \alpha)(x - \beta)(x - \gamma)$$

Hence, we can find formulas for a, b, c in terms of these roots

$$= (x^2 - (\alpha + \beta)x + \alpha\beta)(x - \gamma)$$
$$= x^3 + (-\alpha - \beta - \gamma)x^2 + (\alpha\beta + \alpha\gamma + \beta\gamma)x - \alpha\beta\gamma$$
$$\Rightarrow a = -\alpha - \beta - \gamma$$
$$\Rightarrow b = \alpha\beta + \alpha\gamma + \beta\gamma$$
$$\Rightarrow c = -\alpha\beta\gamma$$

Hence, for roots $\alpha^3, \beta^3, \gamma^3$, we have these corresponding a', b', c' values

$$a' = -\alpha^3 - \beta^3 - \gamma^3$$

$$b' = \alpha^3\beta^3 + \alpha^3\gamma^3 + \beta^3\gamma^3$$
$$c' = -\alpha^3\beta^3\gamma^3$$

We want to express each of these using the original a, b, c. First, notice that

$$a^3 = -\alpha^3 - \beta^3 - \gamma^3 - 3\alpha^2\beta - 3\alpha^2\gamma - 3\alpha\beta^2 - 3\alpha\gamma^2 - 3\beta^2\gamma - 3\beta\gamma^2 - 6\alpha\beta\gamma$$
$$ab = -\alpha^2\beta - \alpha\beta^2 - \alpha^2\gamma - \alpha\gamma^2 - \beta^2\gamma - \beta\gamma^2 - 3\alpha\beta\gamma$$
$$b^3 = \alpha^3\beta^3 + \alpha^3\gamma^3 + \beta^3\gamma^3$$
$$+ 3\alpha^3\beta^2\gamma + 3\alpha^3\beta\gamma^2 + 3\alpha^2\beta^3\gamma + 3\alpha^2\beta\gamma^3 + 3\alpha\beta^3\gamma^2 + 3\alpha\beta^2\gamma^3 + 6\alpha^2\beta^2\gamma^2$$
$$abc = \alpha^3\beta^2\gamma + \alpha^3\beta\gamma^2 + \alpha^2\beta^3\gamma + \alpha^2\beta\gamma^3 + \alpha\beta^3\gamma^2 + \alpha\beta^2\gamma^3 + 3\alpha^2\beta^2\gamma^2$$

Hence,

$$a' = a^3 - 3ab + 3c$$
$$b' = b^3 - 3abc + 3c^2$$
$$c' = c^3$$

Thus our cubic is

$$y = x^3 + (a^3 - 3ab + 3c)x^2 + (b^3 - 3abc + 3c^2)x + c^3$$

A4

A line can only intersect another line at most once; otherwise, they are parallel, which clearly doesn't work if we want our line to intersect all four lines.

Our four points are $(a, 1, 0)$, $(0, b, 1)$, $(1, 0, c)$, $\left(d, d, -\frac{d}{6}\right)$. Since two points specify a line, we need to fix two points such that the line passes through the remaining two. Using the first two points, we have the vector $(-a, b-1, 1)$ and we can choose the first point $(a, 1, 0)$ as origin. Thus, the line's parametric equation is

$$L = (-a, b-1, 1)t + (a, 1, 0)$$

We want to intersect the third line, hence we equate

$$(1, 0, c) = (-a, b-1, 1)t + (a, 1, 0)$$

Thus, $t = c$, from which we derive

$$1 = -ac + a \quad \Rightarrow \quad a = \frac{1}{1-c}$$

$$0 = (b-1)c + 1 \quad \Rightarrow \quad b = 1 - \frac{1}{c}$$

Our line is refined

$$L = \left(-\frac{1}{1-c}, -\frac{1}{c}, 1\right)t + \left(\frac{1}{1-c}, 1, 0\right)$$

To intersect the fourth line, we equate again

$$\left(d, d, -\frac{d}{6}\right) = \left(-\frac{1}{1-c}, -\frac{1}{c}, 1\right) t + \left(\frac{1}{1-c}, 1, 0\right)$$

Thus, $t = -\frac{d}{6}$, from which we derive

$$d = \frac{1 + \frac{d}{6}}{1-c}$$

$$d = 1 + \frac{d}{6c}$$

$$\Rightarrow c = \frac{-1 \pm 5}{12}$$

i.e. $c = -\frac{1}{2}$ or $c = \frac{1}{3}$. The corresponding lines are

$$L = \left(-\frac{2}{3}, 2, 1\right) t + \left(\frac{2}{3}, 1, 0\right) = (-2, 6, 3)\, t + \left(\frac{2}{3}, 1, 0\right)$$

$$L = \left(-\frac{3}{2}, -3, 1\right) t + \left(\frac{3}{2}, 1, 0\right) = (-3, -6, 2)\, t + \left(\frac{3}{2}, 1, 0\right)$$

Chapter 3

Putnam 1985

A1

For each integer 1 to 10, it must be contained in at least one of A_1, A_2, A_3, but it must not be contained in all three. Thus, for each integer, there are 6 choices of belonging: only A_1, only A_2, only A_3, both A_1 and A_2, both A_1 and A_3, both A_2 and A_3. Since there are 10 integers, there are 6^{10} choices of sets, which can be expressed as $2^{10}3^{10}$.

B1

Developing the expression, we get

$$x^5 - \left(\sum_{i=1}^{5} m_i\right) x^4 + \left(\sum_{i,j=1}^{5} m_i m_j\right) x^3$$

$$- \left(\sum_{i,j,k=1}^{5} m_i m_j m_k\right) x^2 + \left(\sum_{i,j,k,l=1}^{5} m_i m_j m_k m_l\right) x - m_1 m_2 m_3 m_4 m_5$$

where $m_i \neq m_j \neq m_k \neq m_l$ in each sum.

We show at least one of the last two coefficients must be nonzero. If $m_1 m_2 m_3 m_4 m_5 = 0$, then one of the terms must be 0. Since multiplication in the original expression is interchangeable, then the order of m_i does not matter. So let $m_1 = 0$. It follows that $\sum_{i,j,k,l=1}^{5} m_i m_j m_k m_l = m_2 m_3 m_4 m_5$, which is the coefficient of x. If this coefficient is 0, then one of m_2, m_3, m_4, m_5 must be 0, which contradicts the fact that each term must be different. Hence, at most one of the coefficients of x^1 and x^0 can be 0, never both. Thus, at least one must be nonzero.

Now, we show at least one of the first two nontrivial coefficients must be nonzero. If the coefficient of x^4 is zero, i.e. $\sum_{i=1}^{5} m_i = 0$, then squaring both

sides we get $\sum_{i=1}^{5} m_i^2 + \sum_{i,j=1}^{5} m_i m_j = 0$. Notice that the second sum is equal to the coefficient of x^3. If we additionally want it to be 0, then we are left with $\sum_{i=1}^{5} m_i^2 = 0$, which is only possible with each $m_i = 0$ since each square is non-negative, which is a contradiction since each m_i must be different. Hence, at most one of the coefficients of x_4 and x_3 can be 0, never both. Thus, at least one must be nonzero.

Since the coefficient of x^5 cannot be influenced by m_i and remains 1, and since we deduced at least two additional nonzero terms, it follows that $k \geq 3$ at least. In fact, we show $k = 3$ by example. Let $m_1 = 0, m_2 = 1, m_3 = -1, m_4 = 2, m_5 = -2$.

$$\begin{aligned} & x(x-1)(x+1)(x-2)(x+2) \\ &= x(x^2-1)(x^2-4) \\ &= x(x^4-5x^2+4) \\ &= x^5 - 5x^3 + 4x \end{aligned}$$

which has three nonzero coefficients. Hence, $k = 3$.

Chapter 4

Putnam 1994

A1

Here, the sum notation doesn't include the upper endpoint,

$$\sum_{i=2^n}^{2^{n+1}} a_i \leq \sum_{i=2^n}^{2^{n+1}} (a_{2i} + a_{2i+1}) = \sum_{i=2^{n+1}}^{2^{n+2}} a_i$$

Since $a_1 > 0$, for the base case $i = 0$, we have

$$\sum_{i=1}^{2} a_i = a_1 > 0$$

Hence, every such sum $\sum_{i=2^j}^{2^{j+1}} a_i$ is greater or equal to $a_1 > 0$ for all j. Now, since

$$\sum_{i=1}^{\infty} a_i = \sum_{j=0}^{\infty} \sum_{i=2^j}^{2^{j+1}} a_i \geq \sum_{j=0}^{\infty} a_1$$

with the latter sum diverging to ∞, it follows that the original sum diverges as well.

A3

Let's draw the triangle ABC with the right angle C north, and the hypothenuse AB horizontal. Suppose for contradiction that it is possible to fill ABC such that same color points are always less than $2 - \sqrt{2}$ distance from each other.

Let A have Color A, and B have Color B, since they are obviously too far from each other ($\sqrt{2} > 2 - \sqrt{2}$, because $\sqrt{2} \approx 1.4$ and $2 - \sqrt{2} \approx 0.6$).

Trace two circles P and Q of radius $2 - \sqrt{2}$ around A and B respectively. Let E be the intersection point of P and AB, and give it Color E. The ratio

$n = \frac{EB}{AB}$ is equal to $n = \frac{\sqrt{2}-(2-\sqrt{2})}{\sqrt{2}} = 2-\sqrt{2}$. Dropping a perpendicular from E to BC, we form an isosceles right triangle EFB at F, with $FB = n \cdot 1 = 2-\sqrt{2}$ where we used the similar triangles ratio n. Hence, $EF = 2-\sqrt{2}$ as well.

Thus, F must be a different color from E, call it Color F. Now, C, being even farther from E than F, thus has a different color from E. It's also too far from A and B, hence it must have the only remaining Color F.

Let G be intersection between Q and AB. By symmetry, G is also too far from C, just like E, and it's also too far from A and B, hence it must have the only remaining Color E. By symmetry, dropping a perpendicular from G to AC, and calling the base point H, we know H must have a different color than G, so by elimination, it must have Color C.

Finally, notice that CFH is an isosceles right triangle. With sides $1-(2-\sqrt{2}) = \sqrt{2}-1$, its hypothenuse FH is equal to $\sqrt{2}(\sqrt{2}-1) = 2-\sqrt{2}$, which is exactly the out-of-bound length between same color points. Since F and H share a same color, but are at a distance of $2-\sqrt{2}$, it's a contradiction, hence the original assumption is wrong. Hence, there exists two points of the same color at a distance at least $2-\sqrt{2}$ for any possible coloring of ABC.

A4

Since A and B have integer entries, then $A+nB$ have integer entries as well.

The inverse of a matrix

$$A = \begin{pmatrix} P & Q \\ R & S \end{pmatrix}$$

is given by

$$A^{-1} = \frac{1}{\det A}\begin{pmatrix} S & -Q \\ -R & P \end{pmatrix}$$

Now, since A is integer entries, it follows that $\det A = PS - QR$ must be an integer. The determinant of the inverse if given by $\det A^{-1} = \frac{1}{\det A}$. Obviously, $\det A \neq 0$. But also, $|\det A| \leq 1$; otherwise, $\det A^{-1}$ would be a fraction (not an integer). Since by assumption, the entries of A^{-1} are integers, then its determinant must also be an integer, so it's impossible for $\det A^{-1}$ to be a fraction. Thus, either $\det A = 1$ or -1.

For similar reasons, either $\det A + nB = 1$ or -1. Now, let

$$B = \begin{pmatrix} W & X \\ Y & Z \end{pmatrix}$$

The determinants are

$$\det A + nB = (P+nW)(S+nZ) + (Q+nX)(R+xY)$$

$$= PS + nPZ + nWS + n^2WZ - QR - nQY - nXR - n^2XY = \pm 1$$

where n ranges from 0 to 4. Taking the difference between successive equations, we get

$$PZ + WS + (2m+1)WZ - QY - XR - (2m+1)XY = \pm 2 \text{ or } 0$$

where it's a known fact that the difference between successive squares are the odd numbers. Here, m ranges from 0 to 3, for a total of 4 equations.

Notice that this is an arithmetic sequence where $2WZ - 2XY$ is the step size. However, there is only 3 possible values: $-2, 0, 2$. The step size cannot be 1, otherwise we have odd values, which don't correspond to the allowed values. It also cannot be greater or equal to 3, because the range is $2-(-2) = 4$, so two steps would already overshoot past the range of values (starting from $2 \cdot 3 = 6$). Hence, $2WZ - 2XY = 0$. Thus, every $A + nB$ share a same determinant of either 1 or -1.

Now, $\det A + 5B$ is simply the next term in this arithmetic sequence, which is a constant 1 or -1. Hence, it's invertible.

Using the same reasoning as before, $A + 5B$ has integer entries, and since its determinant is 1 or -1, then its inverse, which only consists of swapping components in the matrix and dividing by the determinant (which is 1 or -1), is obviously integer entries as well. To be most explicit, integer divided by 1 or -1 either preserves or flips its sign, so the components remain integers.

A5

Consider the set S_1 of one-term sums, which is basically the original terms of (r_n). Since the latter converges to 0, then for every interval (a, b), consider the interval $(x, b) \subseteq (a, b)$ for which $[x, b] \cap \{0\} = \varnothing$ (if $b = 0$, then choose (a, x)). S_1 must contain a finite number of one-term sums; otherwise, (r_n) doesn't converge (distance between terms remains larger than $|x|$). Since (x, b) (or (a, x)) contains only a finite amount of one-term sums, label the latter in increasing order $s_1, \ldots, s_n$, and take the interval $(c, d) = (s_1, s_2)$ to be your answer.

Now, suppose that for the set S_n of n-term sums, for every interval (a, b), there exists (c, d) that does not intersect S_n. Consider the set $S_{n+1} = \{s_\alpha + r_i \mid s_\alpha \in S_n, r_i \in (r_n)\}$.

B1

The first few squares are

$$1, 4, 9, 16, 25, 36, 49, 64, 81, 100$$

$$121, 144, 169, 196, 225, 256, 289, 324, 361, 400$$

$$441, 484, 529, 576, 625$$

For integer 1, the squares within 250 are 1 through 225, which is exactly 15. As we increase up to 5, the squares still range from 1 to 225, where $5+250 = 255$ falls just short of 1 to reach 256.

As we reach 6, the number of squares just keeps increasing, since the lower bound always includes 1 while the upper bound keeps gulping more squares. Every number from 6 up to 251 includes squares from 1 to 225, plus any additional ones gulped by the upper bound, hence the number of squares is greater than the initial 15.

At 252, we finally lose the square 1.

TOO TEDIOUS.

B2

We simply need

$$x^4 + 9x^3 + cx^2 + 9x + 4$$

to have 3 nonzero concavities, hence 2 inflection points.

Taking 2nd derivative, we get

$$12x^2 + 54x + 2c = 0$$

and we want two solutions (for two inflection points), hence

$$54^2 - 4 \cdot 12 \cdot 2c > 0$$

$$\Rightarrow c < \frac{3^5}{8}$$

Now, we show that 2 inflection points is the only requirement. The desired line actually passes through these two inflection points, so we already have two distinct intersection points.

Looking at the original quartic equation, the dominant term x^4 tends to ∞ toward both sides. When the graph dips below the line when we go left from the left inflection points, since it tends to ∞, then by IVT, the graph must intersect the line again. Likewise for the right part of the graph.

A quartic can only have 2 inflection points max (by simple inspection of the quadratic equation for finding the inflection points). So the only other cases to conside are 1 and 0 inflection points.

Chapter 5

Putnam 2000

A2

There are infinitely many odd squares, i.e. there exists infinitely many y such that $y^2 = 2x + 1$. Notice that x is unique for every y^2, since $x_0 \neq x_1 \Rightarrow 2x_0 \neq 2x_1 \Rightarrow 2x_0 + 1 \neq 2x_1 + 1$. For each choice of y^2, let us form three integers:

$$x^2 + 0^2$$

$$x^2 + 1^2$$

$$y^2 + (x-1)^2$$

Let $n = x^2$. Obviously, $x^2 + 1^2 = n + 1$. Now,

$$y^2 + (x-1)^2 = 2x + 1 + (x-1)^2 = x^2 + 2 = n + 2$$

Hence, these are three consecutive sums of two squares. Since there are infinitely many choices of y^2, each associated with a unique choice of x, each associated with a unique choice of n, it follows that there are infinitely many choices of n such that this property holds.

Chapter 6

Putnam 2002

A2

Let five distinct points on the sphere be labelled a, b, c, d, e. Two non-antipodal points uniquely define a great circle, which cut the sphere into two hemisphere.

Consider the case where a, b are non-antipodal. For the remaining c, d, e, each lies either on one closed hemisphere, or its complement, which is an open hemisphere. By pigeon hole principle, there must be one hemisphere with at least two points among c, d, e. If it's the closed hemisphere, then adding points a, b gives it at least four points, satisfying the condition. If it's the open hemisphere, since its closure contains the great circle, we can likewise add points a, b to give at least four points on this closed version.

The remaining case is when a, b are antipodal. Point c, being distinct, must be non-antipodal to both. Hence, a, b, c define a great circle. For the remaining d, e, each lies either on one closed hemisphere, or its complement, which is an open hemisphere. By pigeon hole principle, there must be one hemisphere with at least one point among d, e. If it's the closed hemisphere, adding a, b, c gives at least four points, satisfies the condition. If it's the open hemisphere, since its closure contains the great circle, we can likewise add a, b, c to give at least four points, satisfying the condition.

Thus, for any five distinct points on a sphere, we can find four contained in a same closed hemisphere.

Chapter 7

Putnam 2008

A1

For every triple of real numbers, we have

$$f(x,y) + f(y,z) + f(z,x) = 0$$

Since (x,x,x) is a valid triple form, substituting, we get

$$3f(x,x) = 0$$

Hence, $f(x,x) = 0$ for every real number x.

Using triples of the form (x,x,z), we get

$$f(x,x) + f(x,z) + f(z,x) = 0$$

Since $f(x,x) = 0$, it follows that

$$f(x,z) + f(z,x) = 0$$

$$\Rightarrow f(x,z) = -f(z,x)$$

for every pair of real numbers x, z.

Using this equation, the first equation can be converted to

$$f(x,y) + f(y,z) - f(x,z) = 0$$

$$\Rightarrow f(x,y) - f(x,z) = -f(y,z)$$

Let $g(t) = -f(0,t)$. Computation shows that

$$g(s) - g(t)$$

$$= -f(0,s) + f(0,t)$$

$$= f(0,t) - f(0,s)$$

Using the modified first equation, we get

$$= -f(t, s)$$

$$= f(s, t)$$

Hence, by example, such a function g exists.

A2

Tile the matrix completely using horizontal dominoes of height 1 and length 2. This is possible because 2008 is even, so we can divide the matrix's length by 2. Height is trivially divisible by 1. Hence, this tiling is possible.

Whenever Alan places a number x into one of the cells of a domino, Barbara copies x into the other cell of the same domino. This is possible because Alan starts first, hence there is always an odd amount of numbers in the matrix on Barbara's turn, and since each domino's volume is 2, there must exist a domino with only one number, which Barbara can use.

When the matrix filled, consider its first and second column. These two column vectors are identical, hence the determinant of the entire matrix is zero.

Chapter 8

Putnam 2010

A1

Every integer must be contained in a box. Hence, one of the boxes must contain n. Thus, the agreed sum should satisfy $s \geq n$.

The sum of integers from 1 to n is given by $\frac{n(n+1)}{2}$. Partitioning them into k boxes of sum s, the total sum should be the same, i.e. $sk = \frac{n(n+1)}{2}$.

If n is even, suppose for contradiction that $s = n$. Then

$$sk = \frac{n(n+1)}{2}$$

$$\Rightarrow nk = \frac{n(n+1)}{2}$$

$$\Rightarrow k = \frac{n+1}{2}$$

which is not an integer, hence $s = n$ is impossible for n even.

The next candidate is $s = n + 1$. Now, a pairing exists where 1 is paired with n, and 2 is paired with $n - 1$, and so on. Since there is an even number of integers, the pairing is possible. We have a total of $k = \frac{n}{2}$ pairs (boxes).

To show that this is the largest choice of k, recall that $s \geq n + 1$ and $sk = \frac{n(n+1)}{2}$. If $k > \frac{n}{2}$, then $sk > (n+1)\frac{n}{2}$, contradicting $sk = \frac{n(n+1)}{2}$. Thus, $k \leq \frac{n}{2}$.

If n is odd, let $s = n$. Let n sit in its own box, and pair 1 with $n - 1$, and 2 with $n - 2$, and so on. Since $n - 1$ is even, this pairing is possible. We have a total of $k = \frac{n-1}{2} + 1$ boxes.

To show that this is the largest choice of k, again, if $k > \frac{n+1}{2} - 1$, then $sk > n\left(\frac{n-1}{2} + 1\right) = \frac{n^2-n}{2} + \frac{2n}{2} = \frac{n^2+n}{2} = \frac{n(n+1)}{2}$, contradicting $sk = \frac{n(n+1)}{2}$. Thus, $k \leq \frac{n-1}{2} + 1$.

In conclusion, for n even, $k = \frac{n}{2}$, and for n off, $k = \frac{n-1}{2} + 1$.